

Particle Kinematics: the Move Graphs and the Edge Census

Theorem monograph T02 of the chess-dynamics-lab program

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Keywords: move graph, double counting, connectivity, bipartiteness

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Setting

Each piece on the empty board generates an undirected move graph G_π : the vertices are the 64 squares, and an edge joins two squares whenever a piece of type π can move from one to the other in a single move. This graph is the kinematic passport of the particle: it determines which trajectories are available, what the mobility is, and how the reachability field is shaped. In the “pieces as particles” model the move graph plays the role of the table of admissible impulses: every move is a discrete impulse carrying the particle from a vertex to a vertex.

The task of this monograph is to compute the edge census of all five graph types (rook, bishop, knight, king, queen) in closed form, to prove the connectivity and bipartiteness of the knight graph, to prove the two-component structure of the bishop graph, and to establish the mutual consistency of all numbers with each other and with the mobility theorem T03. All values are verified by protocol C2 and coincide across the seven language implementations of the polyglot core.

Theorem

Theorem T02 (edge census and structure of the move graphs). On the empty board 8×8 the move graphs have exactly the following numbers of edges: rook — 448, bishop — 280, knight — 168, king — 210, queen — 728. The knight graph is connected and bipartite (the parts are the square colours); the bishop graph has exactly two connected components of 32 vertices each — precisely the colour classes; the rook, king and queen graphs are connected.

Proof

One method runs through all the counts — double counting: the number of edges equals half the sum of the vertex degrees, and the degree sum is the total number of directed moves. For the rook the degree of every vertex is 14 (7 squares along the file plus 7 along the rank), so the number of edges is $64 \cdot 14/2 = 448$. For the king the degrees are 3 at the four corners, 5 at the 24 non-corner edge squares and 8 at the 36 interior squares; the degree sum is $4 \cdot 3 + 24 \cdot 5 + 36 \cdot 8 = 12 + 120 + 288 = 420$, hence $420/2 = 210$ edges.

For the bishop we use the closed mobility formula proved in T03: a bishop at (f, r) attacks exactly $14 - |f - r| - |f + r - 7|$ squares — the lengths of the two diagonals through the square minus the square itself on each of them. Summing over all squares: $\sum_{f,r} |f - r| = 2 \sum_{d=1}^7 d(8 - d) = 2 \cdot 84 = 168$, and by symmetry $\sum_{f,r} |f + r - 7| = 168$. The degree sum is therefore $64 \cdot 14 - 168 - 168 = 896 - 336 = 560$, and the number of edges is $560/2 = 280$. A bishop move preserves the square colour (each diagonal step changes both coordinates by amounts of equal parity), so the bishop graph splits into at least two components — the colour classes — of 32 vertices each. Connectivity of each class: from any square of a colour a bishop reaches in one move one of the four central squares of its colour ($d4, e5$ in one class; $e4, d5$ in the other), and the central squares of one colour are pairwise connected through a common diagonal; hence any two squares of a colour are joined in at most three moves, and there are exactly two components.

For the knight the degrees are given by a table: 2 at the four corners, 3 and 4 on the edge, 6 in the diagonally-adjacent-to-corner squares, 8 in the central zone. The full degree sum is 336, giving $336/2 = 168$ edges; the table itself is enumerated and frozen in the protocol, each value checked by direct enumeration of the eight offsets. Bipartiteness: every knight step changes $f + r$ by the odd quantity $(\pm 1) + (\pm 2)$, so the square colour flips with every move, and the colour split is a proper 2-colouring with no monochromatic edge. Connectivity of the knight graph is verified by the protocol with a breadth-first pass: all 64 vertices are reachable; analytically this is the classical fact that a knight joins any two squares in at most five moves (confirmed by the BFS diameter census of the protocol).

For the queen the move graph is the union of the rook and bishop graphs: files, ranks and diagonals from one square. The edge sets intersect trivially: no rook edge is diagonal and no bishop edge is straight. Therefore $|E_Q| = |E_R| + |E_B| = 448 + 280 = 728$. The connectivity of the queen graph is inherited from the rook graph. The theorem is proved.

Protocol data

Move graph census (check C2)

Piece	Directed moves	Edges	Components
Rook	896	448	1
Bishop	560	280	2 (32 squares each)
Knight	336	168	1, bipartite
King	420	210	1
Queen	1456	728	1

Summary of what is proved

We proved the complete edge census of the five kinematic graphs in closed forms, established the bipartiteness of the knight graph with the explicit colour 2-colouring, proved the two-component structure of the bishop graph with the exact identification of the components and the colour classes, and the connectivity of the three remaining graphs. The mechanism is double counting reinforced by closed diagonal-length formulas; all numbers are pairwise consistent (the knight degree sum equals twice the edge census, the queen census splits into rook and bishop).

Verification

Check: `python3 dynamics.py -run C2` or the C2 line of the polyglot core. The edge number is computed two independent ways — enumerating square pairs with a move test and summing degrees; the agreement of both with the tabulated values yields PASS. Bipartiteness is checked by a breadth-first 2-colouring, the components by a full connectivity pass. Reproduction is mandatory in all seven languages of the polyglot core.